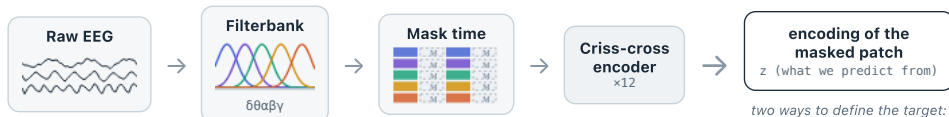
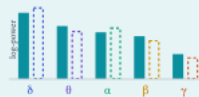


What does the model predict for a masked patch — and why band power, not the band signal?



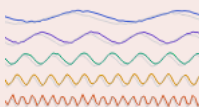
OURS Predict band power



5 numbers / patch
(one log-power per band)
target = **fixed** filterbank power
(frozen, cannot collapse)

✓ A **lossy, denoised summary**: keeps **which** bands carry energy — exactly what clinical labels are defined by — and throws away phase and fine detail. The encoder must learn discriminative **spectral structure**, not noise.

ALTERNATIVE Reconstruct the band signal



$\sim 1280 \times 5$ samples / patch
(every point of every band)
target = full band-domain
waveforms

✗ This is **reconstruction**: to match the waveform the model must fit **phase, fine spectral shape, and noise** — most of which is task-irrelevant. Capacity is spent modelling noise (the failure mode of raw-signal reconstruction, e.g., CBraMod).

Band-power prediction is only the **pretext task**. After pretraining we **discard the prediction head** and keep the **encoder** — that encoder is the foundation model that transfers to downstream tasks (sleep staging, depression, mental workload). We never reconstruct anything at inference.